Testing the Integer Crossing Hypothesis: Does $\alpha^{-1}(\mu) = 137$ at Any Energy Scale?

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Abstract

We test the **Integer Crossing Hypothesis**: the conjecture that there exists an energy scale μ^* such that the running fine structure constant inverse $\alpha^{-1}(\mu^*) = 137$ exactly. This hypothesis is motivated by the algebraic fact that $D^4 - 2D^3 + 2D + 1 = 137$ at $D = 4$ and the empirical observation that $\alpha^{-1}(m_e) = 137.036$ (0.026% from 137). Using Standard Model renormalization group running, we demonstrate that this hypothesis is **falsified**: $\alpha^{-1}(\mu)$ monotonically decreases from 137.036 at $\mu = m_e$ and never crosses 137 at any physical scale. We also rule out cosmological time variation as an explanation, as the required drift exceeds observational bounds by a factor of ~ 26. The proximity of α^{-1} to the integer 137 remains an unexplained coincidence.

1 Introduction

The fine structure constant $\alpha \approx 1/137$ has fascinated physicists since its identification as a fundamental dimensionless parameter of electromagnetism. The fact that $\alpha^{-1} \approx 137.036$ is tantalizingly close to the integer 137 has motivated numerous attempts to derive this value from first principles.

In previous work, we established the algebraic identity:

$$D^4 - 2D^3 + 2D + 1 = 137 \quad \text{(exactly, for } D = 4) \tag{1}$$

This raises a natural question: Is there a physical mechanism that would predict $\alpha^{-1} = 137$ exactly? One possibility is that the running coupling $\alpha^{-1}(\mu)$ crosses 137 at some special energy scale μ^*.

2 The Integer Crossing Hypothesis

Conjecture (Integer Crossing Hypothesis): *There exists an energy scale μ^* such that*

$$\alpha^{-1}(\mu^*) = 137 \quad \text{(exactly)} \tag{2}$$

Motivation:

- $D^4 - 2D^3 + 2D + 1 = 137$ (algebraic fact, $D = 4$)
- $\alpha^{-1}(m_e) = 137.035999084 \pm 0.000000021$ (CODATA 2022)
- Deviation from 137: only 0.026%

3 QED Running Coupling

In quantum electrodynamics, the fine structure constant runs with energy scale μ according to the renormalization group equation. At one-loop:

$$\alpha^{-1}(\mu) = \alpha^{-1}(\mu_0) - \frac{\beta_0}{2\pi} \ln\left(\frac{\mu}{\mu_0}\right) \tag{3}$$

where

$$\beta_0 = \frac{4}{3} \sum_f Q_f^2 N_c \tag{4}$$

is the QED beta function coefficient, summed over active fermions with electric charge Q_f and color multiplicity N_c.

3.1 Key Physical Features

1. **Below** m_e: No charged fermions are active, so $\beta_0 \approx 0$. The coupling is essentially constant.
2. **Above** m_e: Electron loops contribute $\beta_0 = 4/3$, causing α^{-1} to *decrease* with increasing μ.
3. **Direction**: $\alpha^{-1}(\mu)$ monotonically decreases as μ increases due to vacuum polarization screening.

3.2 Measured Values

Scale	μ	$\alpha^{-1}(\mu)$
Electron mass	$m_e = 0.511$ MeV	137.036
Z boson mass	$M_Z = 91.2$ GeV	127.95

4 Analysis: Where Does $\alpha^{-1} = 137$?

4.1 High Energy Direction

Since α^{-1} decreases with increasing μ, and $\alpha^{-1}(m_e) = 137.036 > 137$, there is no scale $\mu > m_e$ where $\alpha^{-1} = 137$.

4.2 Low Energy Direction

To find $\alpha^{-1} = 137$, we would need $\mu < m_e$. However, below the electron mass threshold, the electron decouples from the running:

$$\beta_0 \to 0 \quad \text{for } \mu < m_e \tag{5}$$

This means α^{-1} is essentially **constant** below m_e, frozen at approximately 137.036.

4.3 Conclusion

$$\boxed{\text{There is no scale } \mu \text{ where } \alpha^{-1}(\mu) = 137 \text{ in standard QED.}} \tag{6}$$

5 Full Standard Model Running

We verify this conclusion using the full Standard Model running with all particle thresholds:

Scale	μ (GeV)	$\alpha^{-1}(\mu)$
m_e	0.000511	137.036
m_μ	0.106	135.90
m_τ	1.78	133.47
m_b	4.18	132.30
M_W	80.4	128.07
M_Z	91.2	127.95
m_t	173	127.50

Observation: $\alpha^{-1}(\mu)$ monotonically decreases from 137.036. It never equals 137 at any physical scale.

6 Alternative: Cosmological Time Variation

An alternative hypothesis is that α^{-1} *was* exactly 137 at some cosmological epoch and has drifted to 137.036 over cosmic time.

6.1 Required Drift

$$\frac{\Delta\alpha}{\alpha} = \frac{137.036 - 137}{137} = 2.63 \times 10^{-4} \tag{7}$$

6.2 Observational Constraints

Measurements from quasar absorption spectra constrain time variation of α over cosmological timescales:

$$\left|\frac{\Delta\alpha}{\alpha}\right| < 10^{-5} \quad \text{(at redshift } z \sim 0.5 - 3.5) \tag{8}$$

6.3 Conclusion

The required drift (2.63×10^{-4}) exceeds the observational bound (10^{-5}) by a factor of ~ 26.

$$\boxed{\text{Cosmological drift hypothesis is RULED OUT by current observations.}} \tag{9}$$

7 Summary of Results

7.1 Hypothesis Tested

Integer Crossing Hypothesis: $\exists \mu^*$ such that $\alpha^{-1}(\mu^*) = 137$ exactly.

7.2 Result

FALSIFIED within the Standard Model

7.3 Reasons

1. $\alpha^{-1}(\mu)$ runs from 137.036 at m_e *downward* to ~ 128 at M_Z
2. Below m_e, running stops ($\beta_0 \to 0$), freezing $\alpha^{-1} \approx 137.036$
3. No energy scale exists where $\alpha^{-1} = 137$ exactly
4. Cosmological time variation is ruled out by quasar observations

8 What Remains Unexplained

8.1 Established Facts

1. $D^4 - 2D^3 + 2D + 1 = 137$ (algebraic identity, $D = 4$)
2. $\alpha^{-1}(m_e) = 137.036$ (measured)
3. Deviation: 0.026%

8.2 Open Questions

1. Why is α^{-1} so close to the integer 137?
2. Is there a deeper connection to spacetime dimension $D = 4$?
3. What physics beyond the Standard Model could predict $\alpha^{-1} = 137$?

9 Alternative Conjectures for Future Work

9.1 Conjecture A: UV Integer Quantization

The "bare" coupling at the UV cutoff (Planck scale) might be exactly 137, with radiative corrections producing the observed 137.036.

Test: Compute $\alpha^{-1}(M_{Pl})$ in various BSM scenarios.

9.2 Conjecture B: Dimension-Coupling Connection

The $D = 4$ polynomial identity suggests a deep connection between spacetime dimension and electromagnetic coupling.

Test: Derive a theoretical mechanism linking D and α.

9.3 Conjecture C: Anthropic Selection

α^{-1} being near 137 might reflect anthropic constraints on chemistry and nuclear physics.

Test: Compute sensitivity of complex chemistry to α variations.

10 Conclusion

We have tested the Integer Crossing Hypothesis using Standard Model physics and found it to be **falsified**. The running coupling $\alpha^{-1}(\mu)$ never equals 137 at any physical energy scale, and cosmological time variation cannot account for the 0.026% deviation.

However, the proximity of α^{-1} to the integer 137 remains unexplained. The algebraic connection $D^4 - 2D^3 + 2D + 1 = 137$ at $D = 4$ is a tantalizing hint that may point toward new physics.

This work demonstrates the value of **systematic conjecture testing**: even a falsified hypothesis sharpens our understanding and points toward new directions.

$$\textit{Conjecture} \rightarrow \textit{Test} \rightarrow \textit{Falsify} \rightarrow \textit{Refine} \rightarrow \textit{Repeat}$$

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